

Lateef - Developer information

Welcome font developers!

We welcome other developers who wish to get involved in supporting and enhancing these fonts or who want to modify them.

Permissions granted by the OFL

SIL's fonts are licensed according to the terms of the [SIL Open Font License](#). The OFL allows the fonts to be used, studied, modified and redistributed freely as long as they are not sold by themselves. For details see the OFL.txt and OFL-FAQ.txt files in the package.

Building the fonts from source code

Font sources are published in a [Github project](#). The build process requires [smith](#) and project build parameters are set in the [wscript](#).

Font sources are in the [UFO3](#) format with font family structures defined using [designspace](#). OpenType source code is stored in the [.fea](#) format in the UFO (features.fea) but is maintained in a separate file using the more efficient and powerful [.feax](#) format.

The fonts are built using a completely free and open source workflow using industry-standard tools ([fonttools](#)), a package of custom python scripts ([pysilfont](#)), and a build and packaging system ([Smith](#)). The whole toolchain is available as a Docker container.

Full instructions for setting up the tools and building SIL fonts are available on a dedicated web site: [Building and Modifying SIL Fonts](#). Additional developer information specific to SIL's Arabic fonts can be found at [font-arab-tools README](#).

In addition, much of the code for Scheherazade New, Harmattan, and Lateef is shared. Carefully review the [font-arab-tools developer](#) documentation to see how the code is shared.

Building

The Lateef project can be built from source using [smith](#). This is done via the sequence:

```
smith distclean
smith configure
smith build
smith alltests
```

This project implements two additional `smith build` options that are useful during development:

- `--regOnly` -- build only the Regular weight instead of all weights
- `--norename` -- keep the working names for glyphs rather than change them to production names

Adding characters

After adding glyphs (other than used only as components for building other glyphs) to the font, the following files will also need updating:

- `glyph_data.csv` -- used to set glyph orders and psnames in the built font
- `classes.xml` -- used to define classes used by OpenType. Note that some of the classes defined therein are noted to be "automatically generated" -- these will be updated (from `glyph_data.csv`) the next time `./preflight` is run.
- `opentype/*.feax` -- modify as needed to add needed OpenType behavior
- `tests/*.ftml` -- see below

Generated test files

After adding characters or additional behaviors to the font, test files should be created or enhanced to test the new behaviors. The test files:

- `tests/AllChars-auto.ftml`
- `tests/ALsorted-auto.ftml`
- `tests/DaggerAlef-auto.ftml`
- `tests/DiacTest1-auto.ftml`
- `tests/DiacTest1-short-auto.ftml`
- `tests/FeatLang-auto.ftml`
- `tests/Kern-auto.ftml`
- `tests/KernDigits-auto.ftml`
- `tests/SubtendingMarks-auto.ftml`
- `tests/Yehbarree-auto.ftml`

are generated automatically using `tools/genftmlfiles.sh`. This script, in turn, calls `tools/absngenftml.py` to create each test file. A lot of test generation logic is driven by Unicode character properties and the `glyph_data.csv` file, but sometimes `absngenftml.py` itself needs to be enhanced.

For more information about testing, see [font-arab-tools testing](#).

Contributing to the project

We warmly welcome contributions to the fonts, such as new glyphs, enhanced smart font code, or bug fixes. The [brief overview of contributing changes](#) is a good place to begin. The next step is to contact us by responding to an existing issue or creating an issue in the Github repository and expressing your interest. We can then work together to plan and integrate your contributions.

To enable us to accept contributions in a way that honors your contribution and respects your copyright while preserving long-term flexibility for open source licensing, you would also need to agree to the **SIL Global Contributor License Agreement for Font Software (v1.0)** prior to sending us your contribution. To read more about this requirement and find out how to submit the required form, please visit the [CLA information page](#).

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